

Unit 1 - Math Fundamentals Functions

MSDS 520

Meharry Medical College

Outline

- 1 Why Functions?
- 2 From Relations to Functions
- 3 Domain and Range
- 4 Families of Functions
- 5 Combining Functions
- 6 Inverse Functions
- 7 Functions as Models
- 8 Review

Why Functions?

A Problem to Motivate Us

Mixing an IV bag

“We need a 500 mL bag of saline at a concentration of 4 mg/mL. How many mg of drug go in?”

The rule is one you already know — mass is concentration times volume:

$$m = c \cdot V$$

Fix the bag at $V = 500$ mL and that rule becomes a machine: feed it a target concentration, it returns the mass to weigh out.

$$m(c) = 500c$$

That machine is a **function**. Running it forward ($c = 4 \Rightarrow m = 2000$) is **evaluation**; running it backward (“the vial holds 1500 mg — what concentration will that give?”) is the **inverse**. Both are in this lecture.

Where This Shows Up in Data Science

Almost everything you will build is a function

- **A model is a function:** it takes a patient's measurements and returns a prediction.
- **Every data transformation is a function:** rescaling a column, taking a log, converting units.
- **Error is a function:** “how wrong is this model?” is a number that depends on the model's settings.
- **Training is a search over functions:** pick a family, then pick the best member of it.
- **Reading a result backward** — “what input would have given me this output?” — is an inverse.

Unit 1-2 built **sets**. A function is what connects two sets: it sends each element of one to an element of the other.

What We Will Build

The plan

- ① **Relations**, and what makes one a **function**.
- ② **Domain** and **range** — which inputs are valid, which outputs occur.
- ③ The **families**: polynomial, exponential, logarithmic, trigonometric.
- ④ **Combining** functions — especially **composition**.
- ⑤ **Inverses** — undoing a function, and when that is possible.
- ⑥ Functions as **models**.

Notebooks for this lecture: U1-3_Functions-1_DomainRangeAndFamilies, U1-3_Functions-2_Composition, U1-3_Functions-3_Inverse.

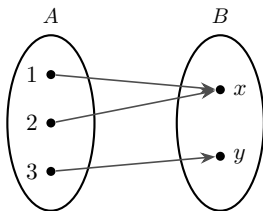
From Relations to Functions

Pairing Things Up

Unit 1-2 ended with **ordered pairs**: (a, b) , where order matters, so $(1, 2) \neq (2, 1)$.

Relation

A **relation** R from a set A to a set B is any set of ordered pairs (a, b) with $a \in A$ and $b \in B$. If $(a, b) \in R$ we say “ a is related to b .”



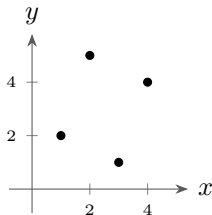
A relation is just a **set of links**. Drawing one arrow per pair turns the list into a picture:

$$R = \{(1, x), (2, x), (3, y)\}$$

Nothing here is required to be tidy — an element of A may link to many elements of B , to one, or to none at all.

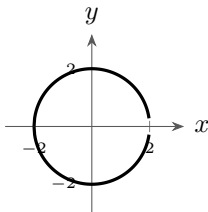
Relations Live in the Plane

When A and B are both $\mathbb{R}$, a pair (x, y) is a **point**, so a relation is a **picture**: every set of points you can draw is a relation.



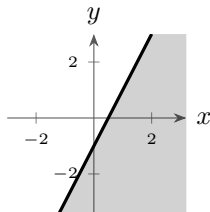
a handful of points

$$R = \{(1, 2), (2, 5), \\ (3, 1), (4, 4)\}$$



a curve

$$R = \{(x, y) \mid x^2 + y^2 = 4\}$$



a whole region

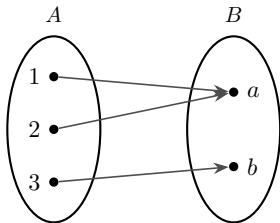
$$R = \{(x, y) \mid y \leq 2x - 1\}$$

All three are perfectly good relations — a finite list, a curve, a shaded half-plane. Only one of them will turn out to be a **function**.

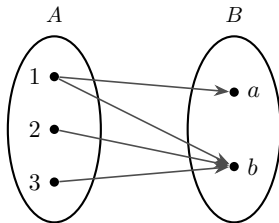
When Is a Relation a Function?

Function

A **function** f from A to B is a relation in which **every** element of A is related to **exactly one** element of B .



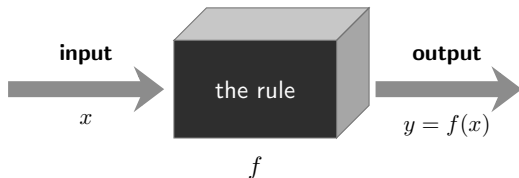
Is a function:
one arrow out of each input



Not a function:
1 has two arrows out

Two inputs sharing one output is fine (**many-to-one**); one input with two outputs is not (**one-to-many**). A relation failing that test is still a relation — just not a function.

A Function Is a Machine



Feed in an x , the rule runs, one y comes out. Two demands hide inside that “exactly one”:

- **At least one:** no input is left without an output — feed the machine a valid input and something must come out.
- **At most one:** no input produces two different outputs — feed it the same input twice and you get the same answer twice.

The IV bag rule $m(c) = 500c$ is exactly this: a box you drop a concentration into and read a mass out of. The rest of the lecture asks three questions about the box — what may go **in**, what comes **out**, and can we run it **backward**?

Notation

Reading $f : A \rightarrow B$

“ f is a function from A to B .” Here A is the **domain** — the valid inputs — and the outputs land in B .

If $f(x) = y$, then x is the **input** (or **argument**) and y is the **output** — the **value of f at x** .

A distinction worth keeping straight

f is the **rule**. $f(x)$ is the **number** that rule produces at x .

So “ $f = x^2$ ” is sloppy; “ $f(x) = x^2$ ” is right. Functions are usually named f , g , h — but any name works, and in code they get real names like `normalize` or `predict`.

The Same Function, Four Ways

Take $f(x) = 2x + 1$ on all of $\mathbb{R}$. The same function can be written down four ways:

As a formula	$f(x) = 2x + 1$			compact and exact	
As a set of pairs	$\{ (x, 2x + 1) \mid x \in \mathbb{R} \}$			every pair, not a list	
As a table	x	0	1	2	a sample of pairs
	$f(x)$	1	3	5	
As a graph	the line through all the pairs				the whole picture

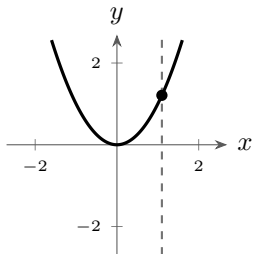
Only the table is incomplete. The domain here is all of $\mathbb{R}$, so the function really pairs *every* real with its output — including $f(0.5) = 2$ and $f(-\pi) = 1 - 2\pi$. A table can only ever show finitely many rows, which is why we still write the formula.

That gap is the job: real data arrives as a **table**, and modeling is guessing the formula behind it.

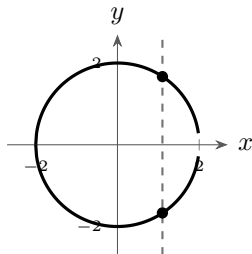
The Vertical Line Test

The graph of f is a picture of the pairs $(x, f(x))$. So “one output per input” has a visual meaning: **no vertical line may cross the graph twice**.

$y = x^2$ — a function



$x^2 + y^2 = 4$ — not a function



The circle fails: that one input meets **two** outputs, so “ y as a function of x ” is not defined here. The circle is a perfectly good *relation* — just not a function.

Exercise: Function or Not?

Try it — which of these are functions from A to B ?

- ① $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (2, 4), (3, 5)\}$
- ② $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (1, 5), (2, 4), (3, 5)\}$
- ③ $A = \{1, 2, 3\}$, $B = \{4, 5\}$, $R = \{(1, 4), (3, 5)\}$
- ④ Each patient $\mapsto$ that patient's date of birth.
- ⑤ Each date of birth $\mapsto$ the patient born on it.
- ⑥ $y^2 = x$, taking x as the input and y as the output.

For each one that fails, say *which* of the two demands it breaks.

Exercise: Solution

Answers

- ① **Yes.** All three inputs are used, one arrow each. (4 being hit twice is fine — many-to-one.)
- ② **No.** 1 is paired with both 4 and 5: “at most one” fails.
- ③ **No.** 2 has no output at all: “at least one” fails.
- ④ **Yes.** A person has exactly one date of birth.
- ⑤ **No.** Many patients share a birthday, so one date leads to many patients — one-to-many.
- ⑥ **No.** The input $x = 4$ gives $y = 2$ **and** $y = -2$. This is the sideways parabola, and it fails the vertical line test.

Compare #4 and #5: reversing a function's arrows does **not** always give a function back. That single observation is the whole story of **inverses**, at the end of this lecture.

Domain and Range

Domain and Range

Definitions

For a function f ,

- the **domain** is the set of every input f accepts;
- the **range** is $\{ f(x) \mid x \text{ in the domain} \}$ — the outputs that actually **occur**.

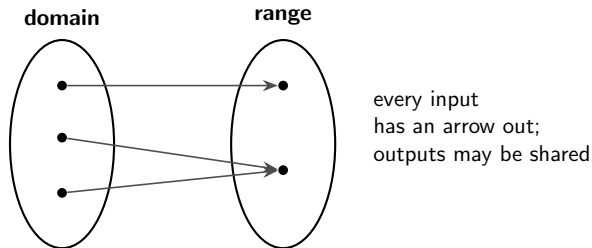
Two different questions, and the second is usually the harder one: the domain asks *what may go in*, the range asks *what comes out*.

Example

$g(x) = x^2$ accepts every real number, so its domain is $\mathbb{R}$. But no square is negative, so its range is only $[0, \infty)$ — the entire negative half of the number line is never produced.

Every input is used; not every value need be produced.

Domain and Range, in a Picture



The domain is **chosen** — it is part of saying what the function is. The range is **discovered** — you find it by asking what the rule can actually produce. That is why a domain is usually read off the formula, while a range is read off the graph.

What Restricts a Domain

Often the domain is “all real numbers.” Three operations spoil that, and they are the ones to watch for:

The three usual suspects

- ① **Division:** $\frac{1}{x}$ requires $x \neq 0$. Division by zero is undefined.
- ② **Even roots:** $\sqrt{x}$ requires $x \geq 0$. No real number squares to a negative.
- ③ **Logarithms:** $\log(x)$ requires $x > 0$. Strictly positive — $\log(0)$ is undefined too.

And the context can restrict it further

A dose cannot be negative; a count cannot be a fraction; an age above 130 is a data-entry error. The **mathematical** domain is what the formula tolerates; the **practical** domain is usually smaller.

Finding a Domain: The Recipe

Procedure

- ① Find every division, even root, and logarithm in the formula.
- ② Write the condition each one forces: denominator $\neq 0$, radicand ≥ 0 , log argument > 0 .
- ③ Solve each condition for x .
- ④ **Intersect** the results — all conditions must hold at once.
- ⑤ Write the answer in interval notation.

Step 4 is where Unit 1-2 pays off: “all conditions at once” is the **AND** of Unit 1-1, which is the **intersection** $\cap$ of the sets of allowed inputs.

Worked Examples

$$f(x) = \sqrt{x - 2}$$

Domain: the root needs $x - 2 \geq 0$, so $x \geq 2$: $[2, \infty)$.

Range: a square root returns non-negative values, and $x - 2$ runs over all of $[0, \infty)$, so every non-negative output is achieved: $[0, \infty)$.

$$g(x) = \frac{1}{x - 3}$$

Domain: need $x - 3 \neq 0$, so $\mathbb{R} \setminus \{3\} = (-\infty, 3) \cup (3, \infty)$.

Range: $1/(x - 3)$ can be made as close to 0 as you like but never equals it: $\mathbb{R} \setminus \{0\}$.

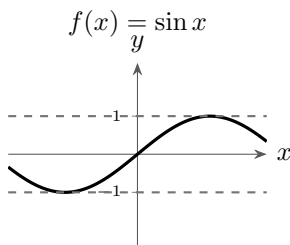
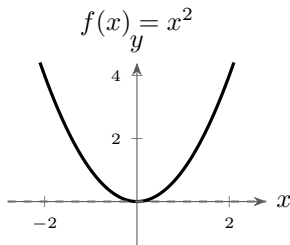
$$h(x) = \ln(x + 1)$$

Domain: need $x + 1 > 0$, so $x > -1$: $(-1, \infty)$.

Range: the logarithm climbs without bound and falls without bound: $\mathbb{R}$.

Range Is the Harder Half

A domain comes from the formula. A range usually comes from **looking at the graph** and asking: how high does it go, how low, and does it skip anything in between?



Left: the parabola bottoms out at 0 and rises forever, so its range is $[0, \infty)$.

Right: the sine wave is trapped between the two dashed lines, so its range is $[-1, 1]$ — a **bounded** function.

Exercise: Domain and Range

Try it — give the domain of each, in interval notation

$$\textcircled{1} \quad f(x) = \sqrt{5 - x}$$

$$\textcircled{2} \quad g(x) = \frac{1}{x^2 - 4}$$

$$\textcircled{3} \quad h(x) = \ln(3 - x)$$

$$\textcircled{4} \quad k(x) = \frac{\sqrt{x}}{x - 1}$$

(careful — this one has **two** conditions)

Then give the **range** of #1 and #3.

Exercise: Solution

Domains

- ① $5 - x \geq 0 \Rightarrow x \leq 5$: $(-\infty, 5]$.
- ② $x^2 - 4 \neq 0 \Rightarrow x \neq \pm 2$: $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$.
- ③ $3 - x > 0 \Rightarrow x < 3$: $(-\infty, 3)$. (Strict — the log excludes its endpoint.)
- ④ The root needs $x \geq 0$; the denominator needs $x \neq 1$. Intersecting:
 $[0, 1) \cup (1, \infty)$.

Ranges

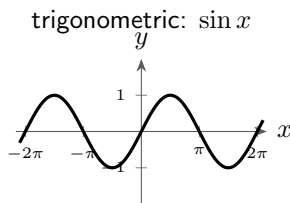
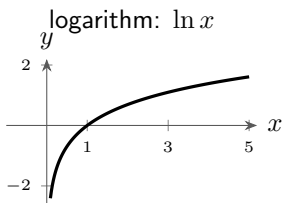
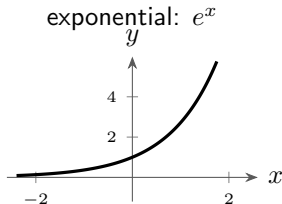
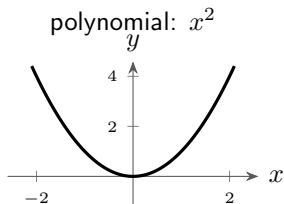
#1: a square root is never negative and $5 - x$ covers all of $[0, \infty)$, so the range is $[0, \infty)$.

#3: as $x \rightarrow 3^-$ the argument $3 - x$ shrinks toward 0 and the log dives without bound; as $x \rightarrow -\infty$ it climbs without bound. Range: $\mathbb{R}$.

#2 is worth a second look: a **squared** expression in the denominator excludes *two* points, not one.

Families of Functions

Four Shapes Worth Memorizing



Curves up in a bowl; shoots up; rises slowly and only for $x > 0$; repeats forever.
Recognizing these four at a glance is the goal — real data keeps producing them.

Linear Functions

Definition

$$f(x) = mx + b$$

m is the **slope** (rise over run — how much f changes per unit of x) and b is the **intercept** (the value at $x = 0$).

- Domain $\mathbb{R}$; range $\mathbb{R}$ when $m \neq 0$, and just $\{b\}$ when $m = 0$ (a flat line).
- **Constant rate of change:** every extra unit of x adds the same m to the output.
- This is the entire content of **linear regression**: choose m and b so the line sits as close as possible to the data.

Slope carries units and therefore meaning: “blood pressure rises 0.4 mmHg per additional kg” *is* a slope.

Polynomials

Definition

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

The largest exponent n is the **degree**; the numbers a_i are the **coefficients**.

Degree	Name	Shape
0	constant	flat line
1	linear	straight line
2	quadratic	one bowl (parabola)
3	cubic	one S-bend

Domain is always $\mathbb{R}$ — polynomials only add and multiply, so nothing is ever undefined. A degree- n polynomial bends at most $n - 1$ times, which is why raising the degree lets a curve chase data more closely — and, as Unit 3 will show, chase the *noise* too.

Exponential Functions

Definition

$$f(x) = a b^x, \quad b > 0, b \neq 1$$

a is the starting value at $x = 0$, and b is the **growth factor** per unit step.

- $b > 1$: **growth** — each step **multiplies** by b . $b < 1$: **decay**.
- Domain $\mathbb{R}$; range $(0, \infty)$ for $a > 0$ — an exponential never reaches 0.
- The natural base is $e \approx 2.718$, and e^x is the default choice throughout math and machine learning.

Linear vs. exponential, in one sentence

A linear function **adds** the same amount each step; an exponential **multiplies** by the same factor each step.

Drug clearance, bacterial growth, and compound interest are all exponential.

Logarithmic Functions

Definition — the log answers “what exponent?”

$$\log_b(y) = x \quad \text{means exactly} \quad b^x = y$$

So $\log_2(8) = 3$, because $2^3 = 8$. The logarithm **undoes** the exponential — our first real example of an inverse.

- Domain $(0, \infty)$; range $\mathbb{R}$ — exactly the exponential's domain and range, swapped.
- Common bases: $b = e$ (written $\ln$), $b = 10$, and $b = 2$ (used in Unit 1-1's information measures).
- The rules that make logs useful: $\log(uv) = \log u + \log v$,
 $\log(u^p) = p \log u$.

That first rule is why we **log-transform** skewed data: it turns multiplication into addition, pulling a long right tail back into a manageable range.

Trigonometric Functions

Definition

$\sin x$ and $\cos x$ are the **periodic** functions: they repeat, forever, with period 2π .

$$\sin(x + 2\pi) = \sin(x)$$

- Domain $\mathbb{R}$; range $[-1, 1]$ — both are **bounded**.
- In $A \sin(\omega x)$: A is the **amplitude** (how tall) and ω controls the **frequency** (how often it repeats).
- $\cos$ is $\sin$ shifted left by $\pi/2$ — the same wave, started at a different point.

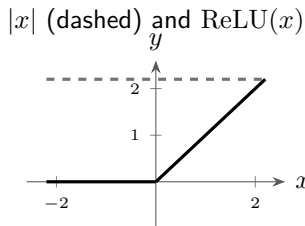
Anything cyclic is modeled this way: circadian rhythm, seasonal disease incidence, the daily swing in an hourly energy series. When a residual plot ripples, a periodic term is usually missing.

Piecewise Functions

A function may be defined by **different rules on different parts of its domain**, so long as each input still gets exactly one output.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

$$\text{ReLU}(x) = \begin{cases} x, & x > 0 \\ 0, & x \leq 0 \end{cases}$$



ReLU — “keep it if positive, otherwise zero” — is the workhorse activation function inside neural networks. Note the **corner** at the origin: piecewise definitions produce kinks, and kinks will matter once we take derivatives in Unit 3.

Transformations: One Family, Many Members

Starting from any $f(x)$, four small edits move and reshape its graph:

Edit	Effect	Direction
$f(x) + c$	shift up by c	vertical, as expected
$f(x + c)$	shift left by c	horizontal, opposite to the sign
$a f(x)$	stretch vertically by a	taller if $a > 1$
$f(-x)$	reflect across the y -axis	left-right flip

The second row is the one that catches people: $f(x + 2)$ moves the graph **left**, not right. The reason is that the input now reaches the value $x + 2$ two units *earlier*.

Every member of a family — every parabola, every sine wave — is one basic shape with these edits applied. Fitting a model is largely a search for the right numbers $a, c, \dots$

Exercise: Families

Try it

- ① Name the family of each: (a) $f(x) = 3^x$, (b) $g(x) = 3x^2 - 1$, (c) $h(x) = \log_{10}(x)$, (d) $k(x) = 4 \sin(2x)$.
- ② Give the domain and range of (a) and (c).
- ③ A colony starts at 200 cells and doubles every hour. Write $N(t)$, the count after t hours.
- ④ Describe how the graph of $(x - 3)^2 + 1$ differs from that of x^2 .

Exercise: Solution

1–2. Families, domains, ranges

- (a) exponential — domain $\mathbb{R}$, range $(0, \infty)$. (b) polynomial (quadratic).
(c) logarithmic — domain $(0, \infty)$, range $\mathbb{R}$. (d) trigonometric, amplitude 4.

3. The colony

Doubling each hour means **multiplying** by 2 each step, so the model is exponential with starting value 200 and growth factor 2:

$$N(t) = 200 \cdot 2^t$$

Check: $N(0) = 200$ and $N(1) = 400$, as required.

4. The transformation

$(x - 3)^2 + 1$ is the parabola shifted **right** 3 and **up** 1 — so its lowest point sits at $(3, 1)$ instead of the origin. Right, not left, because the -3 is inside and horizontal shifts run opposite to their sign.

Combining Functions

The Easy Combinations

Given two functions f and g , we can build new ones point by point:

$$(f + g)(x) = f(x) + g(x), \quad (fg)(x) = f(x)g(x), \quad \left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$$

Watch the domain

The new function is only defined where **both** pieces are — and division adds the condition $g(x) \neq 0$. So

$$\text{domain}(f + g) = \text{domain}(f) \cap \text{domain}(g)$$

another appearance of the intersection from Unit 1-2.

These are useful but not deep. The combination that really matters is the next one.

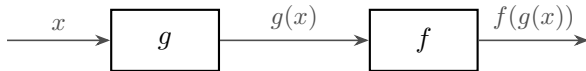
Composition: Feeding One Function into Another

Definition

The **composition** of f with g is

$$(f \circ g)(x) = f(g(x))$$

read “ f of g of x ” or “ f after g .”



Work from the inside out. g runs first, even though f is written first. The parentheses tell you the order, not the reading direction.

A Worked Composition

Let $f(x) = x^2$ and $g(x) = x + 3$.

Evaluate $(f \circ g)(2)$ step by step

$$\begin{aligned}(f \circ g)(2) &= f(g(2)) \\ &= f(2 + 3) \\ &= f(5) \\ &= 5^2 = 25\end{aligned}$$

write it out

inner function first

outer function second

Now as a formula, for every x

$$(f \circ g)(x) = f(x + 3) = (x + 3)^2$$

Substitute the *whole* inner expression wherever the outer function had its input — parentheses included.

Order Matters

With the same $f(x) = x^2$ and $g(x) = x + 3$, swap the order:

	Formula	At $x = 2$
$(f \circ g)(x) = f(g(x))$	$(x + 3)^2$	25
$(g \circ f)(x) = g(f(x))$	$x^2 + 3$	7

Composition is not commutative

In general $f \circ g \neq g \circ f$.

The plain-language version: “add 3, then square” is not the same instruction as “square, then add 3.” Real pipelines have the same hazard — removing outliers and then rescaling gives a different result from rescaling and then removing outliers.

Chains of Steps

Composition is what a **pipeline** is. A typical preprocessing chain is one long composition:

$$x \longrightarrow \underbrace{x - \mu}_{\text{center}} \longrightarrow \underbrace{\frac{x - \mu}{\sigma}}_{\text{scale}} \longrightarrow \underbrace{f\left(\frac{x - \mu}{\sigma}\right)}_{\text{model}}$$

- Each stage is a function; the whole pipeline is their composition.
- Compositions chain to any length: $(f \circ g \circ h)(x) = f(g(h(x)))$ — still inside out.
- A neural network is exactly this: a deep stack of simple functions composed together.

A practical consequence: to apply the pipeline to new data you must re-run **every** stage, in the same order, with the same numbers μ and σ .

Reading a Composition Backward

Just as useful as building compositions is **spotting** them: given a complicated function, name the inside and the outside.

Function	Inner g	Outer f
$\sqrt{x^2 + 1}$	$x^2 + 1$	$\sqrt{\cdot}$
$\ln(3x)$	$3x$	$\ln(\cdot)$
$\sin^2(x)$	$\sin x$	$(\cdot)^2$
e^{-x^2}	$-x^2$	$e(\cdot)$

The domain of a composition needs both stages to be valid: for $\ln(3x)$ the inner value $3x$ must land in the log's domain, forcing $x > 0$.

Hold onto this skill. In Unit 3 the **chain rule** differentiates a composition by handling the inner and outer pieces separately — and it starts by asking exactly this question.

Exercise: Composition

Try it — let $f(x) = 2x - 1$ and $g(x) = x^2$

- ① Compute $(f \circ g)(3)$ and $(g \circ f)(3)$.
- ② Write formulas for $(f \circ g)(x)$ and $(g \circ f)(x)$.
- ③ Let $h(x) = \sqrt{x}$. Give a formula for $(h \circ g \circ f)(x)$ and state its domain.
- ④ Identify an inner and outer function for $\frac{1}{\sqrt{x-4}}$.

Exercise: Solution

1-2

$(f \circ g)(3) = f(9) = 17$, while $(g \circ f)(3) = g(5) = 25$. Different, as expected.

$(f \circ g)(x) = 2x^2 - 1$ and $(g \circ f)(x) = (2x - 1)^2$.

3. A three-stage chain

Inside out: f first, then g , then h :

$$(h \circ g \circ f)(x) = h((2x - 1)^2) = \sqrt{(2x - 1)^2} = |2x - 1|$$

Domain: $(2x - 1)^2$ is never negative, so the root is always valid — the domain is $\mathbb{R}$. The absolute value appears because a square root returns the non-negative root.

4. Decomposition

Inner $g(x) = x - 4$, then $\sqrt{\cdot}$, then $1/(\cdot)$. The domain needs $x - 4 > 0$ — strictly, since that value also sits in a denominator. Domain: $(4, \infty)$.

Inverse Functions

Undoing a Function

Temperature conversion runs both directions:

$$F(c) = \frac{9}{5}c + 32 \quad \text{and back} \quad C(f) = \frac{5}{9}(f - 32)$$

Run one and then the other and you are back where you started:

$$C(F(c)) = c \quad \text{for every } c.$$

Inverse function

If $f : A \rightarrow B$, its **inverse** $f^{-1} : B \rightarrow A$ is the function that undoes it:

$$f^{-1}(f(x)) = x \quad \text{for every } x \text{ in the domain,}$$

$$f(f^{-1}(y)) = y \quad \text{for every } y \text{ in the range.}$$

An inverse reverses every arrow in the mapping diagram. The whole question is whether the result is still a function.

A Warning About the Notation

f^{-1} does **not** mean $1/f$

$$f^{-1}(x) \neq \frac{1}{f(x)}$$

The superscript -1 here means “the undoing function,” not “raised to the power -1 .”

For $f(x) = 3x$:

- the **inverse** is $f^{-1}(x) = x/3$ (divide by three, undoing the multiplication);
- the **reciprocal** is $1/f(x) = 1/(3x)$ — a completely different function.

Unfortunately $\sin^2(x)$ *does* mean $(\sin x)^2$, while $\sin^{-1}(x)$ means the inverse sine. The exponent -1 is the exception, and you simply have to remember it.

When Does an Inverse Exist?

Reversing the arrows can break the “exactly one output” rule — so not every function has an inverse.

One-to-one (injective)

f is **one-to-one** if different inputs always give different outputs:

$$x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

If two inputs shared an output, reversing would send that output to **two** places — one-to-many, not a function.

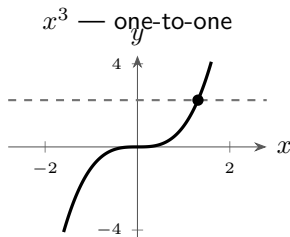
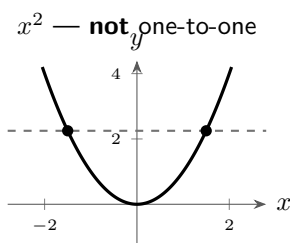
Nothing is left without an output

The other demand takes care of itself: f^{-1} only ever has to answer for values f **actually produces** — its **range** — and every one of those came from some input.

So **one-to-one is exactly the condition**: such an f has an inverse, and that inverse carries the range of f back to its domain.

The Horizontal Line Test

One-to-one has a picture, just as “is a function” did: **no horizontal line may cross the graph twice.**



Squaring loses the sign, so two inputs share each positive output and the square has no inverse on all of $\mathbb{R}$. Restrict the domain to $x \geq 0$ and it becomes one-to-one — which is why $\sqrt{\cdot}$ is defined as the **non-negative** root.

Finding an Inverse by Algebra

Procedure

- ① Write $y = f(x)$.
- ② **Solve for** x in terms of y .
- ③ Swap the names: the expression in y is $f^{-1}(y)$; rename y to x if you prefer.
- ④ Check: does $f^{-1}(f(x))$ simplify to x ?

Example: $f(x) = 3x + 5$

$$y = 3x + 5$$

$$y - 5 = 3x \quad \text{undo the } + 5$$

$$\frac{1}{3}(y - 5) = x \quad \text{undo the } \times 3$$

$$\text{so } f^{-1}(x) = \frac{x - 5}{3}, \quad \text{and } f^{-1}(3x + 5) = \frac{(3x + 5) - 5}{3} = x. \quad \checkmark$$

Undo in the Opposite Order

Notice what the algebra actually did. To build $f(x) = 3x + 5$ you **multiply by 3, then add 5**. To undo it you **subtract 5, then divide by 3** — the opposite operations, in the opposite order.

$$x \xrightarrow{\times 3} 3x \xrightarrow{+5} 3x + 5$$

$$\frac{y - 5}{3} \xleftarrow{\div 3} y - 5 \xleftarrow{-5} y$$

Same idea as taking off a coat and then a sweater: reversing a composition reverses the order of its steps, $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$. It is also why a preprocessing pipeline has to be undone stage by stage, backward, to turn a prediction back into the original units.

Domain and Range Trade Places

Reversing every arrow swaps the roles of the two sets:

$$\text{domain}(f^{-1}) = \text{range}(f), \quad \text{range}(f^{-1}) = \text{domain}(f)$$

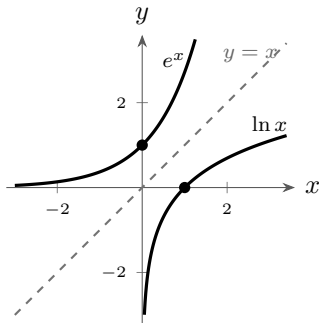
Function	Domain	Range
e^x	$\mathbb{R}$	$(0, \infty)$
$\ln x$	$(0, \infty)$	$\mathbb{R}$

This is the cleanest way to see why $\ln$ refuses non-positive inputs: e^x never *produces* a non-positive value, so $\ln$ is never *asked* about one.

Same story for $\sqrt{\cdot}$ as the inverse of x^2 restricted to $x \geq 0$: the restricted square has range $[0, \infty)$, and that is exactly the square root's domain.

An Inverse Is a Reflection Across $y = x$

Swapping input and output swaps the coordinates of every point: if (a, b) is on the graph of f , then (b, a) is on the graph of f^{-1} . Geometrically, that is a **mirror across the line $y = x$** .



The two marked points are $(0, 1)$ on e^x and $(1, 0)$ on $\ln x$ — the same pair of numbers, in the opposite order.

Exercise: Inverses

Try it

- ① Find the inverse of $f(x) = 4x - 7$, and check it.
- ② Find the inverse of $g(x) = x^3 + 1$.
- ③ Does $h(x) = x^2 - 4$ have an inverse on all of $\mathbb{R}$? If not, restrict its domain so that it does, and give the inverse there.
- ④ Solve $2^x = 32$ for x . (Which function undoes an exponential?)

Exercise: Solution

1-2

$$y = 4x - 7 \Rightarrow x = \frac{y+7}{4}, \text{ so } f^{-1}(x) = \frac{x+7}{4}. \quad \text{Check:}$$

$$f^{-1}(4x-7) = \frac{4x-7+7}{4} = x. \quad \checkmark$$

$$y = x^3 + 1 \Rightarrow x = \sqrt[3]{y-1}, \text{ so } g^{-1}(x) = \sqrt[3]{x-1}. \quad (\text{Cubing is one-to-one, so no restriction is needed.})$$

3. The one that needs a restriction

No: $h(-3) = h(3) = 5$, so h fails the horizontal line test. Restrict to $x \geq 0$; there h is one-to-one, and $y = x^2 - 4 \Rightarrow x = \sqrt{y+4}$, giving $h^{-1}(x) = \sqrt{x+4}$ with domain $[-4, \infty)$ — which is precisely the range of the restricted h .

4. Undoing an exponential

Apply the logarithm base 2: $x = \log_2(32) = 5$, since $2^5 = 32$. Whenever the unknown sits in an **exponent**, the log is the tool that brings it down.

Functions as Models

A Model Is a Function

The setup

We have training data: pairs (x_i, y_i) for $i = 1, \dots, n$, where x_i holds a patient's **features** and y_i is the **label** we want to predict.

Goal: find a function $f : X \rightarrow Y$ with $f(x_i) \approx y_i$ — and, more importantly, with $f(x) \approx y$ for patients we have never seen.

- The **domain** X is the space of valid inputs — which measurements, in which units, within which ranges.
- The **range** Y says what kind of answer comes out: a number (regression), or a label from a finite set (classification).

“Machine learning” is the business of choosing that function from data instead of writing it down by hand.

Inputs vs. Parameters

Look again at the linear function, now with its two roles labelled:

$$f(x) = \underbrace{m}_{\text{parameter}} \underbrace{x}_{\text{input}} + \underbrace{b}_{\text{parameter}}$$

- The **input** x changes from patient to patient.
- The **parameters** m and b are fixed once the model is trained — they identify *which* line, out of the whole family, we are using.

Two decisions, not one

Choosing a family (linear? exponential? something with a bend?) is modeling. **Choosing the parameters within that family** is fitting. The families earlier in this lecture are the menu you are choosing from.

Error Is a Function Too

Once the data is fixed, the quality of a fit depends only on the parameters. That is itself a function — of m and b , not of x :

$$E(m, b) = \sum_{i=1}^n \left(\underbrace{y_i}_{\text{truth}} - \underbrace{(mx_i + b)}_{\text{prediction}} \right)^2$$

- Squaring makes every miss count as positive, so errors cannot cancel out.
- **Training** means finding the m and b that make E as small as possible.
- “Find the input that minimizes a function” is **optimization** — the subject of Unit 3.

Notice the shift in viewpoint: the same symbols that were *unknowns to be estimated* on the previous slide are the *inputs* of this function. That reversal is the central move in all of model fitting.

Functions of Several Variables

Real models rarely take a single number. With p features, the input is a whole list:

$$f(x_1, x_2, \dots, x_p) = w_1x_1 + w_2x_2 + \dots + w_px_p + b$$

- The domain is now a set of **tuples** — exactly the ordered p -tuples from Unit 1-2. One tuple is one row of your data table.
- Everything from this lecture still applies: one output per input, a domain, a range, composition, and sometimes an inverse.
- Unit 2 gives these tuples a name — **vectors** — and a compact algebra for handling many of them at once.

The bookkeeping grows; the concept does not change at all.

Back to the Opening Question

The IV bag, in this lecture's vocabulary

- ① “Mass depends on the target concentration” names a **function**
 $m(c) = 500c$ — one concentration, one mass.
- ② Concentrations we can actually mix form its **domain**; the masses they need, its **range**.
- ③ Doubling the concentration doubles the mass, so the **family** is linear — a straight line through the origin, slope 500.
- ④ “How many mg for 4 mg/mL?” is **evaluation**: compute $m(4) = 2000$.
- ⑤ “The vial holds 1500 mg — what concentration is that?” is the **inverse**:
 $m^{-1}(1500) = \frac{1500}{500} = 3 \text{ mg/mL}$.
- ⑥ If the order is written in grams but the scale reads milligrams, that unit conversion is one more stage in a **composition**.

Six ideas from one clinical sentence — and every one of them is on a slide in this deck.

Review

Review

What we covered

- ① A **relation** is any set of ordered pairs; a **function** pairs *each* input with *exactly one* output.
- ② Many-to-one is allowed, one-to-many is not — the **vertical line test**.
- ③ **Domain** = valid inputs, **range** = the outputs that actually occur. Watch $1/x$, $\sqrt{x}$, $\log x$.
- ④ Four **families** — polynomial, exponential, logarithmic, trigonometric — plus piecewise rules like ReLU, and the **transformations** that shift and stretch them.
- ⑤ **Composition** $(f \circ g)(x) = f(g(x))$ runs inside out, and order matters; a pipeline is a composition.
- ⑥ An **inverse** f^{-1} undoes f ; it exists exactly when f is **one-to-one**, and its graph is the reflection across $y = x$.
- ⑦ A **model** is a function; its **error** is a function of the parameters, and minimizing it is Unit 3.

Notebooks: U1-3_Functions-1_DomainRangeAndFamilies, -2_Composition, -3_Inverse. **Previous:** Unit 1-2, Sets. **Next:** Unit 1-4, Series.